Workflow

The system follows these steps:

1. **Capture Video Input**: The system continuously captures video frames from a camera.
2. **Hand Tracking & Fingertip Detection**: MediaPipe Hand Tracking detects the user's hand and identifies the fingertip position.
3. **Object Detection**: YOLOv8 detects objects in the frame while filtering out objects close to the hand.
4. **Object Selection**: The system determines whether the user is pointing at an object.
5. **Stability Check**: Ensures the user consistently points at an object for a specific duration to avoid false detections.
6. **Audio Feedback**: If an object is confirmed, the system announces its name using text-to-speech.
7. **Real-time Processing**: The system continuously updates detections and audio feedback as the user moves their hand.

System Components

1. Video Capture

The system captures real-time video using OpenCV:

```
cap = cv2.VideoCapture(1)  # Capture video from camera
```

2. Hand Tracking & Fingertip Detection

The HandTracker class utilizes MediaPipe to detect the user's hand and determine if they are pointing at something:

```
self.hands = self.mpHands.Hands(max_num_hands=1, min_detection_confidence=0.7,
min_tracking_confidence=0.6)
```

- Detects fingertips and applies smoothing to stabilize movements.
- Determines if only the index finger is extended while other fingers are folded.

3. Object Detection

The ObjectDetector class uses YOLOv8 for object detection:

```
self.model = YOLO("yolov8n.pt")
```

- Filters objects below a confidence threshold.
- Ignores objects too close to the hand to avoid misinterpretation.
- Applies Non-Maximum Suppression (NMS) to refine detections.

4. Object Selection Logic

The ObjectPointer class maps fingertip coordinates to detected objects:

```
if x1 <= fingertip[0] <= x2 and y1 <= fingertip[1] <= y2:
    return obj  # If fingertip is inside the bounding box
```

- Ensures the object is within the pointing region before confirming selection.

5. Stability Check

To prevent accidental detections, the system ensures the user points at an object for a specific duration before triggering audio feedback:

```
if last_detected_object == label:
    if current_time - last_detection_time >= DETECTION_DELAY:
        audio_feedback.speak(label)
```

6. Audio Feedback

The AudioFeedback class converts detected object names into speech using pyttsx3:

```
self.engine.say(text)
self.engine.runAndWait()
```

- Runs in a separate thread to avoid blocking real-time processing.

Threading & Optimization

To ensure smooth real-time performance, the system uses:

- **Multi-threading** for running object detection separately from video processing.
- **Locks** to synchronize shared resources and prevent race conditions.
- **Exponential Smoothing** to stabilize fingertip movements.

Implementation Flowchart

1. **Start video capture**
2. **Detect hand and identify fingertip**
3. **Run object detection on the frame**
4. **Check if fingertip is pointing at an object**
5. **Verify stability before confirming selection**
6. **Provide audio feedback via text-to-speech**
7. **Repeat steps in a continuous loop**
8. **Exit on user command**